DSA STUDY BLUEPRINT SERIES

65. Previous Smaller Element Optimal Solution

Finding Previous Smaller Elements using Increasing Stacks

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Previous Smaller:** First element to the left that is strictly smaller.
- Maintain index stack in increasing element order.

Memory & Execution Blueprint

Arr: [4, 5, 2] ⇒ Previous Smaller of 5 is **4**. Previous Smaller of 2 is -1.

C++ Implementation Reference

Standard code layout and syntax

```
vector prevSmaller(vector& arr) {
    vector ans(arr.size(), -1);
    stack st;
    for (int i=0; i= arr[i]) st.pop();
        if (!st.empty()) ans[i] = st.top();
        st.push(arr[i]);
    }
    return ans;
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	O(N)
Space Complexity	O(N)

Key Practices & Gotchas

- **Important:** Pop elements while they are greater than or equal to current to maintain strictly increasing stack.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]`:

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	5 > 4 (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.